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Information page

Abstract

There is provided an information page of a security document in which no non-uniform external force acts on an IC module. The information page of a security document includes a core layer, an IC module holding layer provided on one surface of the core layer via an intermediate layer, an IC module held in the IC module holding layer, and a laser printing layer provided on the other surface of the core layer. The intermediate layer and the IC module holding layer are both made of polycarbonate.

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Background/Summary

(1) This Application is a Continuation of U.S. application Ser. No. 16/485,963 filed on Aug. 14, 2019, which is a National Stage Entry of PCT Application No. PCT/JP2018/005351 filed on Feb. 15, 2018, which claims priority from Japanese Application No. 2017-027038 filed on Feb. 16, 2017. The entire contents of these applications are incorporated herein by reference in their entirety.

TECHNICAL FIELD

(1) The present invention relates to an information page of a security document having an IC module, such as a passport, and more particularly to an information page of a security document which can prevent deformation of an IC module.

BACKGROUND ART

(2) A security document such as a passport contains an information page with a built-in IC module.

(3) In general, an information page with a built-in IC module has a multi-layer structure including a

plurality of layers, in which the IC module is incorporated.

(4) Such an information page can be obtained by preparing a multi-layer structure comprised of a laminate of layers, in which an IC module is interposed, and subjecting the multi-layer structure to hot pressing.

(5) However, it is possible that upon the hot pressing of the multi-layer structure, a non-uniform external force may be applied to the IC module due to differences in thermal shrinkage rate between the layers, resulting in deformation of the IC module.

(6) Such deformation of the IC module, caused by the application of a non-uniform external force thereto, can affect the performance of the IC module and, in addition, is undesirable in terms of appearance.

CITATION LIST

Patent Literature

(7) Patent document 1: Published Japanese Translation No. 2009-514708 of the PCT International Publication

SUMMARY OF THE INVENTION

Problems to be Solved by the Invention

(8) The present invention has been made in view of the above situation. It is therefore an object of the present invention to provide an information page of a security document in which no non-uniform external force acts on an IC module, and there is no occurrence of deformation of the IC module.

Means for Solving the Problems

(9) The present invention provides an information page of a security document, comprising: a core layer having flexibility for incorporating the information page into the security document; an IC module holding layer provided on one surface of the core layer via an intermediate layer; and an IC module held in the IC module holding layer and including a substrate, and an IC chip provided on the substrate, wherein the intermediate layer and the IC module holding layer are both made of polycarbonate.

(10) In a preferred embodiment of the present invention, the information page further comprises a laser printing layer provided on the other surface of the core layer.

(11) In a preferred embodiment of the present invention, the IC module holding layer has a multi-layer construction including a substrate accommodating layer accommodating the substrate, and an IC chip accommodating layer accommodating the IC chip, and an antenna, connected to the IC chip of the IC module, is interposed between the substrate accommodating layer and the IC chip accommodating layer.

(12) In a preferred embodiment of the present invention, a hologram is interposed between any two layers of the information page.

(13) In a preferred embodiment of the present invention, the information page further comprises a first external transparent layer provided on the laser printing layer and having print formed on its inner surface.

(14) In a preferred embodiment of the present invention, the information page further comprises a second external transparent layer provided on the IC module holding layer and having print formed on its inner surface.

(15) In a preferred embodiment of the present invention, the information page further comprises a first white layer provided between the core layer and the laser printing layer and having print formed on its outer surface.

(16) In a preferred embodiment of the present invention, the information page further comprises a first external transparent layer provided on the laser printing layer.

(17) In a preferred embodiment of the present invention, the information page further comprises a second white layer provided on the IC module holding layer and having print formed on its outer surface.

(18) In a preferred embodiment of the present invention, the information page further comprises a transparent window made of a transparent material and penetrating at least the core layer and the IC module holding layer.

(19) The present invention also provides a security document incorporating any one of the above-described information pages.

Advantageous Effects of the Invention

(20) According to the present invention, no non-uniform external force acts on an IC module upon hot pressing, and therefore there is no occurrence of deformation of the IC module.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a cross-sectional side view of an information page of a security document according to a first embodiment, showing a cross-section in the line-I direction of FIG. 5;

(2) FIG. 2 is a cross-sectional side view of an IC module;

(3) FIG. 3 is a diagram showing a positional relationship between an antenna and the IC module;

(4) FIG. 4 is a perspective view of a security document;

(5) FIG. 5 is a plan view of the information page;

(6) FIG. 6 is a cross-sectional side view of an information page of a security document according to a second embodiment, showing a cross-section in the line-VI direction of FIG. 5;

(7) FIG. 7 is a cross-sectional side view of an information page of a security document according to a third embodiment, showing a cross-section in the line-VII direction of FIG. 5;

(8) FIG. 8A is a diagram showing an antenna and an IC module, disposed on an intermediate layer and a substrate accommodating layer;

(9) FIG. 8B is a diagram showing the antenna and the IC module when viewed from the back side of the intermediate layer and the substrate accommodating layer;

(10) FIG. 8C is an enlarged plan view of the IC module, showing an enlarged view of the portion C of FIG. 8A; and

(11) FIG. 8D is an enlarged cross-sectional view of the IC module.

DESCRIPTION OF THE EMBODIMENTS

First Embodiment

(12) An information page of a security document according to a first embodiment will now be described in detail with reference to FIGS. 1 through 5 and some specific examples.

(13) At the outset, the security document incorporating the information page according to the first embodiment will now be described with reference to FIGS. 4 and 5.

(14) The security document **1** shown in FIGS. 4 and 5 includes a passport, etc., and contains the information page **10**. The information page **10** is attached, in a connecting portion **10b**, to a cover **2** and other pages **3** of the security document e.g. by sewing.

(15) The information page **10** includes an information portion **10a**, and the connecting portion **10b** for attachment to the cover **2** and to the other pages **3**.

(16) The information portion **10a** of the information page **10** contains personal information such as the name, the date of birth, a photograph, etc. of a passport holder. Part of the information is printed by using, for example, the below-described laser light technique.

(17) As shown in FIG. 5, the connecting portion **10b** of the information page **10** is composed of a core layer **11** having a net-like structure. The core layer **11** will be described below in greater detail. The net-like structure of the connecting portion **10b** is formed of a fabric of crossing yarns each composed of a plurality of single fibers.

(18) Alternatively, the net-like structure may be provided by a method which involves perforating a uniform film to create a desired pattern. What is essential to the present invention is that the net has

a sufficient number of “yarns”, and that it is quite difficult to attach the yarns to each other without leaving a visible trace associated with attempted counterfeiting.

(19) The core layer **11** of the connecting portion **10b** extends into the information portion **10a**.

(20) The information page **10** of the security document will now be further described with reference to FIGS. **1** through **3**. As described above, the information page **10** includes the information portion **10a** and the connecting portion **10b**, and is attached, in the connecting portion **10b**, to the cover **2** and the other pages **3** of the security document **1**, thus constituting the security document **1**.

(21) The information page **10** includes the core layer **11** having flexibility and having a net-like structure, an IC module holding layer **12** provided on the lower surface (one surface) of the core layer **11** via an intermediate layer **13**, an IC module **20** held in the IC module holding layer **12**, and a laser printing layer **15** provided on the upper surface (the other surface) of the core layer **11** via a first white layer **23**.

(22) The core layer **11** having a net-like structure includes a body portion **11a** which is bonded to the IC module holding layer **12**, the laser printing layer **15**, etc. and which, together with them, constitutes the information portion **10a**, and a connecting portion **11b** constituting the connecting portion **10b** which is composed solely of the core layer **11** and in which the information page **10** is attached to the cover **2** and the other pages **3**.

(23) The IC module **20** includes a substrate **21**, and an IC chip **22a** provided on the substrate **21** and covered with a sealing resin **22b**. The IC chip **22a** and the sealing resin **22b** constitute an IC chip body **22** (see FIG. **2**). In the illustrated case, the thickness of the IC chip body **22** is larger than the thickness of the substrate **21**. For example, the thickness of the IC chip body **22** is 50 μm to 220 μm , while the thickness of the substrate **21** is 50 μm to 100 μm . By making the thickness of the IC chip body **22** larger than the thickness of the substrate **21**, deformation of the entire IC module **20** can be reduced.

(24) The IC module **20** having the above construction is held in the IC module holding layer **12**. In the illustrated case, the IC module holding layer **12** includes a substrate accommodating layer **12a** accommodating the substrate **21** of the IC module **20** and housing the substrate **21**, and an IC chip body accommodating layer (also referred to as an IC chip accommodating layer) **12b** accommodating the IC chip body **22** containing the IC chip **22a**, and housing the IC chip body **22**.

(25) An antenna **25**, connected to the IC chip **22a** of the IC module **20**, is disposed between the substrate accommodating layer **12a** and the IC chip accommodating layer **12b** of the IC module holding layer **12**. The antenna **25** is made of copper, and is interposed between the substrate accommodating layer **12a** and the IC chip accommodating layer **12b** (see FIGS. **1** and **3**).

(26) The IC module holding layer **12** has an IC module housing space **40** in which the IC module **20** is housed and which is composed of an opening **38** formed in the substrate accommodating layer **12a**, and an opening **39** formed in the IC chip accommodating layer **12b**. The IC module housing space **40** is closed from above and below with the intermediate layer **13** and with an additional intermediate layer **14** provided on the lower surface of the IC module holding layer **12**.

(27) The laser printing layer **15** is blackened by irradiation with laser light so that a desired character can be displayed. To this end, the laser printing layer **15** is formed of polycarbonate containing a laser coloring accelerator and, when irradiated with laser light, the laser coloring accelerator has heat energy. The heat energy chars and blackens carbon contained in the polycarbonate in contact with the laser coloring accelerator, resulting in display of a character.

(28) A first external transparent layer **17** is provided on the upper surface of the laser printing layer **15**, and a volume hologram **35** is interposed between the laser printing layer **15** and the first external transparent layer **17**. The volume hologram **35** can display, for example, a crown in the information page **10** (see FIGS. **4** and **5**).

(29) Print **30**, such as characters, a pattern, etc., has been formed on the lower surface (inner surface) of the first external transparent layer **17**.

(30) The print **30**, such as characters, a pattern, etc., includes information displayed on the front surface of the information page **10** of the security document. Such information includes, for example, general information other than individual information such as name and address. The individual information such as name and address is formed by irradiating the laser printing layer **15** with laser light.

(31) A second external transparent layer **18** is provided on the lower surface of the additional intermediate layer **14**. Print **30**, such as characters, a pattern, etc., has been formed also on the upper surface (inner surface) of the second external transparent layer **18**.

(32) The first white layer **23** functions to enhance visibility of the print **30**, which has been formed on the lower surface of the first external transparent layer **17**, when viewing the print **30** from the side of the first external transparent layer **17**.

(33) In particular, with the first white layer **23** provided under the first external transparent layer **17**, the first white layer **23** serves as a background and enables the print **30**, which is colored dark and formed on the lower surface of the first external transparent layer **17**, to be easily viewed (checked) from the side of the first external transparent layer **17**.

(34) There is also provided a transparent window **36** made of a transparent material and penetrating the first white layer **23**, the core layer **11**, the intermediate layer **13**, the IC module holding layer **12** and the additional intermediate layer **14**.

(35) The materials of the respective layers constituting the information page **10** will now be described. At the outset, the core layer **11** having a net-like structure is made of PET, nylon, or the like.

(36) The intermediate layer **13**, the first external transparent layer **17** and the second external transparent layer **18** are all made of transparent polycarbonate.

(37) The substrate accommodating layer **12a** and the IC chip accommodating layer **12b** of the IC module holding layer **12**, the first white layer **23** and the additional intermediate layer **14** are all made of white polycarbonate.

(38) The laser printing layer **15** is made of polycarbonate containing a laser coloring accelerator.

(39) A method for producing the thus-constructed information page **10** of the security document will now be described.

(40) First, the intermediate layer **13** and the substrate accommodating layer **12a** of the IC module holding layer **12** are laminated together. Subsequently, the substrate **21** of the preformed IC module **20** is inserted into the opening **38** of the substrate accommodating layer **12a**. Subsequently, the antenna **25** is disposed on the substrate accommodating layer **12a**, and connected to the IC chip **22a** of the IC module **20**.

(41) Thereafter, the IC chip accommodating layer **12b** is laminated to the substrate accommodating layer **12a**. The lamination of the IC chip accommodating layer **12b** to the substrate accommodating layer **12a** is performed such that the IC chip body **22** of the IC module **20** is housed in the opening **39** of the IC chip accommodating layer **12b**. Subsequently, the additional intermediate layer **14** is laminated to the IC chip accommodating layer **12b**.

(42) Thereafter, the laminate consisting of the intermediate layer **13**, the substrate accommodating layer **12a** and the IC chip accommodating layer **12b** of the IC module holding layer **12**, and the additional intermediate layer **14** is subjected to hot pressing to heat and press the laminate, thereby bonding the layers of the laminate together.

(43) The IC module housing space **40** in which the IC module **20** is housed is surrounded by the intermediate layer **13**, the substrate accommodating layer **12a**, the IC chip accommodating layer **12b** and the additional intermediate layer **14** which are all made of polycarbonate. Thus, the IC module housing space **40** is surrounded by the polycarbonate materials having approximately the same thermal shrinkage rate. Accordingly, the IC module housing space **40** contracts uniformly with good balance during the hot pressing; therefore, no non-uniform external force will be applied by the interior walls of the IC module housing space **40** to the IC module **20** housed in the IC

module housing space **40**, resulting in no deformation of the IC module **20**. This makes it possible to maintain the performance and the appearance of the IC module **20**.

(44) Next, the core layer **11** and the first white layer **23** are laminated to the laminate consisting of the intermediate layer **13**, the IC module holding layer **12** and the additional intermediate layer **14**, and then the laser printing layer **15** is laminated to the first white layer **23**.

(45) Thereafter, the volume hologram **35** is provided on the laser printing layer **15**, and then the first external transparent layer **17**, having the print **30** on the inner surface, is laminated to the laser printing layer **15**. Subsequently, the second external transparent layer **18**, having the print **30** on the inner surface, is laminated to the additional intermediate layer **14**. The resulting laminate is subjected to hot pressing, thereby obtaining the information page **10** of the security document.

(46) Next, laser light is applied to the laser printing layer **15** of the thus-obtained information page **10** from the side of the first external transparent layer **17** to perform necessary printing on the laser printing layer **15**.

(47) As described hereinabove, according to this embodiment, no non-uniform external force acts on the IC module **20** upon the hot pressing, and therefore there is no occurrence of deformation of the IC module **20**. In particular, as described above, the IC module **20** will not be deformed upon the hot pressing because the intermediate layer **13**, the substrate accommodating layer **12a**, the IC chip accommodating layer **12b** and the additional intermediate layer **14** are all made of polycarbonate. This reduces deformation of the laser printing layer **15** formed over the IC module **20**, and can thereby enhance visibility of print formed.

(48) Though in the above-described embodiment the volume hologram **35** is interposed between the laser printing layer **15** and the first external transparent layer **17**, the volume hologram **35** may be provided at a position corresponding to the transparent window **36** and between the first white layer **23** and the laser printing layer **15** (see the phantom line of FIG. **1**). By providing the volume hologram **35** at such a position corresponding to the transparent window **36**, the volume hologram **35** can be viewed from both the front side and the back side of the information page **10**.

Second Embodiment

(49) An information page of a security document according to a second embodiment will now be described with reference to FIG. **6**.

(50) In the second embodiment shown in FIG. **6**, the same symbols are used for the same components or elements as those of the first embodiment shown in FIGS. **1** through **5**, and a detailed description thereof is omitted.

(51) As shown in FIG. **6**, the information page **10** includes an information portion **10a** and a connecting portion **10b**, and is attached, in the connecting portion **10b**, to a cover **2** and other pages **3** of a security document **1**, thus constituting the security document **1**.

(52) The information page **10** includes a core layer **11** having flexibility and having a net-like structure, an IC module holding layer **12** provided on the lower surface (one surface) of the core layer **11** via an intermediate layer **13**, an IC module **20** held in the IC module holding layer **12**, and a laser printing layer **15** provided on the upper surface (the other surface) of the core layer **11** via a first white layer **33**.

(53) The core layer **11** having a net-like structure includes a body portion **11a** which is bonded to the IC module holding layer **12**, the laser printing layer **15**, etc. and which, together with them, constitutes the information portion **10a**, and a connecting portion **11b** constituting the connecting portion **10b** which is composed solely of the core layer **11** and in which the information page **10** is attached to the cover **2** and the other pages **3**.

(54) The IC module **20** is held in the IC module holding layer **12**. In the illustrated case, the IC module holding layer **12** includes a substrate accommodating layer **12a** accommodating the substrate **21** of the IC module **20** and housing the substrate **21**, and an IC chip body accommodating layer (also referred to as an IC chip accommodating layer) **12b** accommodating the IC chip body **22** containing the IC chip **22a**, and housing the IC chip body **22**.

(55) An antenna **25**, connected to the IC chip **22a** of the IC module **20**, is disposed between the substrate accommodating layer **12a** and the IC chip accommodating layer **12b** of the IC module holding layer **12**. The antenna **25** is made of copper, and provided on a substrate **25a** and, together with the substrate **25a**, is interposed between the substrate accommodating layer **12a** and the IC chip accommodating layer **12b** (see FIG. 6).

(56) The IC module holding layer **12** has an IC module housing space **40** in which the IC module **20** is housed and which is composed of an opening **38** formed in the substrate accommodating layer **12a**, and an opening **39** formed in the IC chip accommodating layer **12b**. The IC module housing space **40** is closed from above and below with the intermediate layer **13** and with an additional intermediate layer **14a** provided on the lower surface of the IC module holding layer **12**.

(57) The laser printing layer **15** is blackened by irradiation with laser light so that a desired character can be displayed. To this end, the laser printing layer **15** is formed of polycarbonate containing a laser coloring accelerator and, when irradiated with laser light, the laser coloring accelerator has heat energy. The heat energy chars and blackens carbon contained in the polycarbonate in contact with the laser coloring accelerator, resulting in display of a character.

(58) A first external transparent layer **17** is provided on the upper surface of the laser printing layer **15**, and a volume hologram **35** is interposed between the laser printing layer **15** and the first external transparent layer **17**. The volume hologram **35** can display, for example, a crown in the information page **10** (see FIGS. 4 and 5).

(59) Print **30**, such as characters, a pattern, etc., has been formed on the upper surface (outer surface) of the first white layer **33**.

(60) A second white layer **34** and a second external transparent layer **18** are provided in this order on the lower surface of the additional intermediate layer **14a**. Print **30**, such as characters, a pattern, etc., has been formed also on the lower surface (outer surface) of the second white layer **34**.

(61) The first white layer **33** functions to enhance visibility of the print **30**, which has been formed on the first white layer **33**, when viewing the print **30** from the side of the first external transparent layer **17**.

(62) In particular, since the print **30**, which is light-colored, is provided on the first white layer **33**, the print **30** can be easily viewed (checked) from the side of the first external transparent layer **17**.

(63) Similarly, light-colored print **30** has been formed on the second white layer **34**. Since the print **30** is provided on the second white layer **34**, the print **30** can be easily viewed (checked) from the side of the second external transparent layer **18**.

(64) There is also provided a transparent window **36** made of a transparent material and penetrating the first white layer **33**, the core layer **11**, the intermediate layer **13**, the IC module holding layer **12**, the additional intermediate layer **14a**, and the second white layer **34**.

(65) The materials of the respective layers constituting the information page **10** will now be described. At the outset, the core layer **11** having a net-like structure is made of PET, nylon, or the like.

(66) The intermediate layer **13**, the first external transparent layer **17**, the additional intermediate layer **14a** and the second external transparent layer **18** are all made of transparent polycarbonate.

(67) The substrate accommodating layer **12a** and the IC chip accommodating layer **12b** of the IC module holding layer **12**, the first white layer **33** and the second white layer **34** are all made of white polycarbonate.

(68) The laser printing layer **15** is made of polycarbonate containing a laser coloring accelerator.

(69) The sizes of the respective layers constituting the information page **10** shown in FIG. 6 will now be described.

(70) The overall thickness of the information page **10** shown in FIG. 6 is, for example, 900 μm . In one example, the thickness of the first external transparent layer **17** is 105 μm , the thickness of the laser printing layer **15** is 50 μm , the thickness of the first white layer **33** is 105 μm , the thickness of the core layer **11** is 130 μm , the thickness of the intermediate layer **13** is 40 μm , the thickness of the

substrate accommodating layer **12a** is 105 μm , the thickness of the IC chip accommodating layer **12b** is 150 μm , the thickness of the additional intermediate layer **14a** is 40 μm , the thickness of the second white layer **34** is 105 μm , and the thickness of the second external transparent layer **18** is 50 μm .

(71) A method for producing the thus-constructed information page **10** of the security document will now be described.

(72) First, the intermediate layer **13** and the substrate accommodating layer **12a** of the IC module holding layer **12** are laminated together. Subsequently, the substrate **21** of the preformed IC module **20** is inserted into the opening **38** of the substrate accommodating layer **12a**.

(73) Subsequently, the antenna **25** is disposed on the substrate accommodating layer **12a**, and connected to the IC chip **22a** of the IC module **20**.

(74) Thereafter, the IC chip accommodating layer **12b** is laminated to the substrate accommodating layer **12a**. The lamination of the IC chip accommodating layer **12b** to the substrate accommodating layer **12a** is performed such that the IC chip body **22** of the IC module **20** is housed in the opening **39** of the IC chip accommodating layer **12b**. Subsequently, the additional intermediate layer **14a** is laminated to the IC chip accommodating layer **12b**.

(75) Thereafter, the laminate consisting of the intermediate layer **13**, the substrate accommodating layer **12a** and the IC chip accommodating layer **12b** of the IC module holding layer **12**, and the additional intermediate layer **14a** is subjected to hot pressing to heat and press the laminate, thereby bonding the layers of the laminate together.

(76) The IC module housing space **40** for housing the IC module **20** is surrounded by the intermediate layer **13**, the substrate accommodating layer **12a**, the IC chip accommodating layer **12b** and the additional intermediate layer **14a** which are all made of polycarbonate. Thus, the IC module housing space **40** is surrounded by the polycarbonate materials having approximately the same thermal shrinkage rate. Accordingly, the IC module housing space **40** contracts uniformly with balance during the hot pressing; therefore, no non-uniform external force will be applied by the interior walls of the IC module housing space **40** to the IC module **20** housed in the IC module housing space **40**, resulting in no deformation of the IC module **20**. This makes it possible to maintain the performance and the appearance of the IC module **20**.

(77) Next, the core layer **11** and the first white layer **33** having the print **30** on the outer surface are laminated to the laminate consisting of the intermediate layer **13**, the IC module holding layer **12** and the additional intermediate layer **14a**, and then the laser printing layer **15** is laminated to the first white layer **33**.

(78) Thereafter, the volume hologram **35** is provided on the laser printing layer **15**, and then the first external transparent layer **17** is laminated to the laser printing layer **15**. Subsequently, the second white layer **34**, having the print **30** on the outer surface, and the second external transparent layer **18** are laminated sequentially to the additional intermediate layer **14a**. The resulting laminate is subjected to hot pressing, thereby obtaining the information page **10** of the security document.

(79) Next, laser light is applied to the laser printing layer **15** of the thus-obtained information page **10** from the side of the first external transparent layer **17** to perform necessary printing on the laser printing layer **15**.

(80) As described hereinabove, according to this embodiment, no non-uniform external force acts on the IC module **20** upon the hot pressing, and therefore there is no occurrence of deformation of the IC module **20**. In particular, as described above, the IC module **20** will not be deformed upon the hot pressing because the intermediate layer **13**, the substrate accommodating layer **12a**, the IC chip accommodating layer **12b** and the additional intermediate layer **14a** are all made of polycarbonate. This reduces deformation of the laser printing layer **15** formed over the IC module **20**, and can thereby enhance visibility of print formed.

(81) Though in the above-described embodiment the volume hologram **35** is interposed between the laser printing layer **15** and the first external transparent layer **17**, the volume hologram **35** may

be provided at a position corresponding to the transparent window **36** and between the first white layer **33** and the laser printing layer **15** (see the phantom line of FIG. **6**). By providing the volume hologram **35** at such a position corresponding to the transparent window **36**, the volume hologram **35** can be viewed from both the front side and the back side of the information page **10**.

Third Embodiment

(82) An information page of a security document according to a third embodiment will now be described with reference to FIGS. **7** through **8D**.

(83) In the third embodiment shown in FIGS. **7** through **8D**, the same symbols are used for the same components or elements as those of the first embodiment shown in FIGS. **1** through **5** or the second embodiment shown in FIG. **6**, and a detailed description thereof is omitted.

(84) As shown in FIG. **7**, the information page **10** includes an information portion **10a** and a connecting portion **10b**, and is attached, in the connecting portion **10b**, to a cover **2** and other pages **3** of a security document **1**, thus constituting the security document **1**.

(85) The information page **10** includes a core layer **11** having flexibility and having a net-like structure, an IC module holding layer **12** provided on the lower surface (one surface) of the core layer **11** via an intermediate layer **13**, an IC module **20** held in the IC module holding layer **12**, and a laser printing layer **15** provided on the upper surface (the other surface) of the core layer **11** via a first white layer **33**.

(86) The core layer **11** having a net-like structure includes a body portion **11a** which is bonded to the IC module holding layer **12**, the laser printing layer **15**, etc. and which, together with them, constitutes the information portion **10a**, and a connecting portion **11b** constituting the connecting portion **10b** which is composed solely of the core layer **11** and in which the information page **10** is attached to the cover **2** and the other pages **3**.

(87) The IC module **20** is held in the IC module holding layer **12**. In this embodiment, the IC module holding layer **12** includes a substrate accommodating layer **12a** accommodating the substrate **21** of the IC module **20** and housing the substrate **21**, and an IC chip body accommodating layer (also referred to as an IC chip accommodating layer) **12b** accommodating the IC chip body **22** containing the IC chip **22a**, and housing the IC chip body **22**.

(88) An antenna **25**, connected to the IC chip **22a** of the IC module **20**, is disposed between the substrate accommodating layer **12a** and the IC chip accommodating layer **12b** of the IC module holding layer **12**. The antenna **25** is made of copper, and provided on a substrate **25a** and, together with the substrate **25a**, is interposed between the substrate accommodating layer **12a** and the IC chip accommodating layer **12b** (see FIG. **7**).

(89) The IC module holding layer **12** has an IC module housing space **40** in which the IC module **20** is housed and which is composed of an opening **38** formed in the substrate accommodating layer **12a**, and an opening **39** formed in the IC chip accommodating layer **12b**. The IC module housing space **40** is closed from above and below with the intermediate layer **13** and with an additional intermediate layer **14a** provided on the lower surface of the IC module holding layer **12**.

(90) The laser printing layer **15** is blackened by irradiation with laser light so that a desired character can be displayed. To this end, the laser printing layer **15** is formed of polycarbonate containing a laser coloring accelerator and, when irradiated with laser light, the laser coloring accelerator has heat energy. The heat energy chars and blackens carbon contained in the polycarbonate in contact with the laser coloring accelerator, resulting in display of a character.

(91) A first external transparent layer **17** is provided on the upper surface of the laser printing layer **15**, and a volume hologram **35** is interposed between the laser printing layer **15** and the first external transparent layer **17**. The volume hologram **35** can display, for example, a crown in the information page **10** (see FIGS. **4** and **5**).

(92) Print **30**, such as characters, a pattern, etc., has been formed on the lower surface (inner surface) of the first external transparent layer **17** and on the upper surface (outer surface) of the first white layer **33**.

(93) A second white layer **34** and a second external transparent layer **18** are provided in this order on the lower surface of the additional intermediate layer **14a**. Print **30**, such as characters, a pattern, etc., has been formed also on the lower surface (outer surface) of the second white layer **34**.

(94) There is also provided a transparent window **36** made of a transparent material and penetrating the first white layer **33**, the core layer **11**, the intermediate layer **13**, the IC module holding layer **12**, the additional intermediate layer **14a**, and the second white layer **34**.

(95) The materials of the respective layers constituting the information page **10** will now be described. At the outset, the core layer **11** having a net-like structure is made of PET, nylon, or the like.

(96) The intermediate layer **13**, the first external transparent layer **17**, the additional intermediate layer **14a** and the second external transparent layer **18** are all made of transparent polycarbonate.

(97) The substrate accommodating layer **12a** and the IC chip accommodating layer **12b** of the IC module holding layer **12**, the first white layer **33** and the second white layer **34** are all made of white polycarbonate.

(98) The laser printing layer **15** is made of polycarbonate containing a laser coloring accelerator.

(99) Of the above layers, the first white layer **33** functions to enhance visibility of the print **30**, which has been formed on the lower surface of the first external transparent layer **17**, when viewing the print **30** from the side of the first external transparent layer **17**.

(100) In particular, with the first white layer **33** provided under the first external transparent layer **17**, the first white layer **33** serves as a background and enables the print **30**, which is colored dark and formed on the lower surface of the first external transparent layer **17**, to be easily viewed (checked) from the side of the first external transparent layer **17**.

(101) Light-colored print **30** has been formed on the first white layer **33**. Since the print **30** is provided on the first white layer **33**, the print **30** can be easily viewed (checked) from the side of the first external transparent layer **17**.

(102) Further, light-colored print **30** has been formed on the second white layer **34**. Since the print **30** is provided on the second white layer **34**, the print **30** can be easily viewed (checked) from the side of the second external transparent layer **18**.

(103) The transparent window **36**, penetrating the first white layer **33**, the core layer **11**, the intermediate layer **13**, the IC module holding layer **12**, the additional intermediate layer **14a**, and the second white layer **34**, will now be described.

(104) The transparent window **36** can be set up by sequentially laminating the first white layer **33**, the core layer **11**, the intermediate layer **13**, the IC module holding layer **12** incorporating the IC module **20**, the additional intermediate layer **14a**, and the second white layer **34**, forming a through-hole in the resulting laminate, and then inserting the transparent window **36** of polycarbonate into the through-hole.

(105) The transparent window **36** of polycarbonate functions to prevent counterfeiting based on the presence/absence of the transparent window **36**. The transparent window **36** may be formed of half portions **36a**, **36a**, and an image film **46** having a particular image and interposed between the portions **36a**, **36a**. In this case, the transparent window **36** having the image film **46** is inserted into the through-hole.

(106) The information page **10** contains the volume hologram **35** (see FIG. 7), including one which displays a crown, provided between the laser printing layer **15** and the first external transparent layer **17**. The volume hologram **35** can be obtained by transferring a transfer sheet, containing the volume hologram **35**, to the upper surface of the laser printing layer **15**. Subsequently, the first external transparent layer **17** is laminated to the laser printing layer **15** to which the volume hologram **35** has been transferred.

(107) It is also possible to interpose the volume hologram **35** between the laser printing layer **15** and the first white layer **33** by first transferring a transfer sheet, containing the volume hologram **35**, to the lower surface of the laser printing layer **15** to thereby provide the volume hologram **35** on

the laser printing layer **15**, and then laminating the first white layer **33** to the laser printing layer **15**. (108) In the above-described embodiments, the antenna **25** is connected to the IC module **20** (see FIG. 3). The antenna **25**, connected to the IC module **20**, may be disposed as shown in FIGS. 8A through 8D.

(109) FIG. 8A is a diagram showing the antenna **25** and the IC module **20**, disposed on the intermediate layer **13** and the substrate accommodating layer **12a**, and FIG. 8B is a diagram showing the antenna **25** and the IC module **20** when viewed from the side of the intermediate layer **13** and the substrate accommodating layer **12a**.

(110) As shown in FIGS. 8A through 8D, the antenna **25** is disposed in a winding shape on a peripheral portion of the intermediate layer **13** and the substrate accommodating layer **12a**, and a part **25a** of the antenna **25** extends inward from the winding and is connected to the IC module **20** provided inside the winding of the antenna **25**.

(111) Referring to FIGS. 8C and 8D, the IC module **20** includes a substrate **21**, and an IC chip **22a** provided on the substrate **21** and covered with a sealing resin **22b**. The IC chip **22a** and the sealing resin **22b** constitute an IC chip body **22**.

(112) While the core layer **11** is disposed over the intermediate layer **13** in the above-described embodiments (see FIGS. 1, 6 and 7), the core layer **11** may be disposed under the additional intermediate layer **14** (see FIG. 1) or under the additional intermediate layer **14a** (see FIGS. 6 and 7).

DESCRIPTION OF THE SYMBOLS

(113) **1** security document **10** information page **10a** information portion **10b** connecting portion **11** core layer **11a** body portion **11b** connecting portion **12** IC module holding layer **12a** substrate accommodating layer **12b** IC chip accommodating layer **13** intermediate layer **14** additional intermediate layer **15** laser printing layer **17** first external transparent layer **18** second external transparent layer **20** IC module **21** substrate **22** IC chip body **22a** IC chip **22b** sealing resin **23** first white layer **25** antenna **30** print **31** lenticular lens **33** first white layer **34** second white layer **35** volume hologram **36** transparent window **40** IC module housing space

Claims

1. An information page of a security document, comprising: a core layer having a net-like structure formed of a fabric of crossing yarns each composed of a plurality of single fibers and the core layer having a body portion and a connecting portion; the body portion of the core layer is bonded directly on one surface of a polycarbonate intermediate layer, and a polycarbonate IC module holding layer is bonded directly on a second surface of the polycarbonate intermediate layer opposite to the one surface; and an IC module embedded in the polycarbonate IC module holding layer, the IC module including a substrate and an IC chip provided on the substrate, wherein the IC module is directly connected on a first side to the polycarbonate intermediate layer and directly connected on a second side to a second polycarbonate intermediate layer, wherein the connecting portion of the core layer protrudes from the page such that it does not touch any other layer of the information page and is the only portion of the information page that is configured to be attached to the security document; wherein an opening is provided in at least the core layer, the polycarbonate intermediate layer, the polycarbonate IC module holding layer, and the second polycarbonate intermediate layer, and wherein a volume hologram is provided above said opening such that the volume hologram is viewable through the opening from opposing sides of the information page, and wherein a transparent window is provided in the opening.
2. The information page of a security document according to claim 1, further comprising a laser printing layer provided on a second surface of the body portion of the core layer.
3. The information page of a security document according to claim 2, further comprising a first external transparent layer provided on the laser printing layer and having print formed on its inner

surface.

4. The information page of a security document according to claim 3, further comprising a second external transparent layer provided on the IC module holding layer and having print formed on its inner surface.

5. The information page of a security document according to claim 2, wherein the IC module holding layer has a multi-layer construction including a substrate accommodating layer accommodating the substrate, and an IC chip accommodating layer accommodating the IC chip, and wherein an antenna, connected to the IC chip of the IC module, is interposed between the substrate accommodating layer and the IC chip accommodating layer.

6. The information page of a security document according to claim 2, wherein a hologram is interposed between any two layers of the information page.

7. The information page of a security document according to claim 2, further comprising a first white layer provided between the core layer and the laser printing layer and having print formed on its outer surface.

8. The information page of a security document according to claim 7, further comprising a first external transparent layer provided on the laser printing layer.

9. The information page of a security document according to claim 2, further comprising a transparent window made of a transparent material and penetrating at least the core layer and the IC module holding layer.

10. A security document incorporating the information page according to claim 2.

11. The information page of a security document according to claim 1, wherein the IC module holding layer has a multi-layer construction including a substrate accommodating layer accommodating the substrate, and an IC chip accommodating layer accommodating the IC chip, and wherein an antenna, connected to the IC chip of the IC module, is interposed between the substrate accommodating layer and the IC chip accommodating layer.

12. The information page of a security document according to claim 11, further comprising a first external transparent layer provided on the laser printing layer and having print formed on its inner surface.

13. The information page of a security document according to claim 12, further comprising a second external transparent layer provided on the IC module holding layer and having print formed on its inner surface.

14. The information page of a security document according to claim 1, wherein a hologram is interposed between any two layers of the information page.

15. The information page of a security document according to claim 1, further comprising a first white layer provided between the core layer and the laser printing layer and having print formed on its outer surface.

16. The information page of a security document according to claim 15, further comprising a first external transparent layer provided on the laser printing layer.

17. The information page of a security document according to claim 16, further comprising a second white layer provided on the IC module holding layer and having print formed on its outer surface.

18. The information page of a security document according to claim 15, further comprising a second white layer provided on the IC module holding layer and having print formed on its outer surface.

19. The information page of a security document according to claim 1, further comprising a transparent window made of a transparent material and penetrating at least the core layer and the IC module holding layer.

20. A security document incorporating the information page according to claim 1.
